

Instructions:

1. Run task_0a.py to generate the vector database if not already generated.
2. Press Run All (or restart kernel and run all cells).
3. You will be prompted to provide input values.

– Task 1: Implement a program which, given (a) a query label, l (b) a user selected feature space, and (c) positive integer k, identifies and visualizes the most relevant k images for the given label l, along with their scores, under the selected feature space.

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In [ ]: LABEL = input("Provide a valid label name / label ID present in the Caltech101 dataset")
FEATURE_SPACE = input("Provide a feature space [color, hog, avgpool, layer3, layer4, layer5, layer5_fc, layer5_fc2, layer5_fc3, layer5_fc4, layer5_fc5, layer5_fc6, layer5_fc7, layer5_fc8, layer5_fc9, layer5_fc10, layer5_fc11, layer5_fc12, layer5_fc13, layer5_fc14, layer5_fc15, layer5_fc16, layer5_fc17, layer5_fc18, layer5_fc19, layer5_fc20, layer5_fc21, layer5_fc22, layer5_fc23, layer5_fc24, layer5_fc25, layer5_fc26, layer5_fc27, layer5_fc28, layer5_fc29, layer5_fc30, layer5_fc31, layer5_fc32, layer5_fc33, layer5_fc34, layer5_fc35, layer5_fc36, layer5_fc37, layer5_fc38, layer5_fc39, layer5_fc40, layer5_fc41, layer5_fc42, layer5_fc43, layer5_fc44, layer5_fc45, layer5_fc46, layer5_fc47, layer5_fc48, layer5_fc49, layer5_fc50, layer5_fc51, layer5_fc52, layer5_fc53, layer5_fc54, layer5_fc55, layer5_fc56, layer5_fc57, layer5_fc58, layer5_fc59, layer5_fc60, layer5_fc61, layer5_fc62, layer5_fc63, layer5_fc64, layer5_fc65, layer5_fc66, layer5_fc67, layer5_fc68, layer5_fc69, layer5_fc70, layer5_fc71, layer5_fc72, layer5_fc73, layer5_fc74, layer5_fc75, layer5_fc76, layer5_fc77, layer5_fc78, layer5_fc79, layer5_fc80, layer5_fc81, layer5_fc82, layer5_fc83, layer5_fc84, layer5_fc85, layer5_fc86, layer5_fc87, layer5_fc88, layer5_fc89, layer5_fc90, layer5_fc91, layer5_fc92, layer5_fc93, layer5_fc94, layer5_fc95, layer5_fc96, layer5_fc97, layer5_fc98, layer5_fc99, layer5_fc100]")
K = int(input("Enter K, the number of similar images K to find for the selected label and feature space"))
```

```
In [ ]: # Find the k most similar images for a given query label for the given feature space
from utils.query_input_processor import check_label_is_valid

LABEL = check_label_is_valid(LABEL)
```

Input label: 100
 Downloading dataset if not present.
 Files already downloaded and verified
 Output label: yin_yang

```
In [ ]: # Load feature space and representative label vectors for that feature space
from utils.database_utils import retrieve

feature_vectors = retrieve(f'{FEATURE_SPACE}.pt')
rep_label_vectors = retrieve(f'rep_label_{FEATURE_SPACE}.pt')
```

```
In [ ]: from utils.dataset_utils import initialize_dataset
dataset = initialize_dataset()

from utils.distance_utils import get_distance_fn, top_k_min_distance_ranker
top_k = top_k_min_distance_ranker(K, rep_label_vectors[LABEL], feature_vectors)

print("\nK =", K, "most similar images by", FEATURE_SPACE, "feature space using", dataset.feature_space)
```

```
for i in range(len(top_k)):
    distance, image_id, label = top_k[i]

    print(str(i + 1) + ".", "\tImage ID: ", image_id, "\t\tDistance: ", distance)
    display(dataset[image_id][0])
```

Downloading dataset if not present.
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K = 10 most similar images by color feature space using distance: cityblock
1. Image ID: 8674 Distance: 17298.172



2. Image ID: 3184

Distance: 18357.35



3. Image ID: 3132

Distance: 18385.219



4. Image ID: 6156

Distance: 19652.059



5. Image ID: 3334

Distance: 19680.377



6. Image ID: 7380

Distance: 19697.662



7. Image ID: 7984

Distance: 19793.059



8. Image ID: 8628

Distance: 19808.84



9. Image ID: 7106

Distance: 19968.738



10. Image ID: 5632

Distance: 20195.375

